Stock Return Prediction

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1. Our Initial Idea

Our project began with the goal of predicting stock returns for water-sector stocks in specific. Our research question was that a stock universe that is specific to a sector would allow us to find drivers unique to water infrastructure and utilities, and build predictive models around that.

Our initial set consisted of 8 stocks: AWK, AWR, BMI, CWT, JCI, VRT, WTRG, and XYL. Alongside this, we constructed a set of 16 predictors combining company fundamentals with market technicals, using data from 2013 to 2026.

Beyond fundamental variables, we also intended to include external macro proxy variables intended to capture broader forces acting on the water sector, for example hyperscaler AI capital expenditure (as a proxy for data-center-driven water demand) and regional drought monitors (as a proxy for water scarcity pressure).

For predictive modelling, we shortlisted three methods: Lasso regression, Partial Least Squares (PLS), and Bagged Regression Splines. These were chosen to cover a range of approaches: a regularized linear model (Lasso), a dimension-reduction technique suited to correlated predictors (PLS), and a flexible non-linear method (Bagged Splines).

Our target variables were defined as quarterly and annual excess returns above the S&P 500 index, meaning the models were not predicting raw returns but returns in excess of the broad market benchmark, a more demanding and arguably more useful prediction target for relative performance.

Predictors used

- 60-Day Volatility
- Dividend Yield
- Asset Growth
- 12-Month Momentum
- Net Income Growth
- Return on Equity
- Debt-to-Equity
- Current Ratio

2. Data Scraping Pipeline

Assembling a reliable dataset required building a multi-source data scraping pipeline, since no single provider offered everything we needed with the correct timing and structure.

We initially considered Macrotrends as a data source because it contained HTML tables that were easy to scrape. However, Macrotrends did not report the filing date of the underlying financial data. Without a filing date, there was no way to know when a given piece of fundamental information actually became public, which introduced look-ahead bias, since a model could inadvertently be trained on information before it was actually available to investors in real time.

Look-ahead bias occurs when a model is trained using information that would not actually have been available at the point in time it is meant to be predicting from. In our case, a company's fundamentals (such as return on equity or debt-to-equity) are only made public once they are filed, not as of the date they describe. If a data point is linked to the period it describes rather than the date it was actually disclosed, a model can effectively be trained on future information relative to the prediction point. This makes backtested performance look artificially strong, since that same information would never have been accessible to an investor trading in real time.

To resolve this, we moved to scraping raw financials directly from the SEC's website (SEC EDGAR), which does report filing dates and therefore allows the timing of each data point to be properly aligned with when it was publicly disclosed.

Price data was sourced separately using the TidyQuant R package, covering the period from 2010 to 2026.

We then wrote an R script to merge the SEC-sourced fundamentals with the TidyQuant-sourced price data and to compute the relevant financial ratios used as predictors.

The pipeline was not without its challenges. One key issue was inconsistent SEC tagging of financial line items across different filings and companies, which risked silently misaligning or mislabeling data during the merge. To address this, we built a dedicated verification script to check and validate the SEC tagging before it was used downstream.

3. Results of Predictive Modelling on the Initial Dataset

With the 8-stock water-sector universe and 16 predictors in place, we trained and evaluated the three shortlisted models: Lasso, Partial Least Squares, and Bagged Regression Splines.

Model	Adjusted R^2	Train RMSE	Test RMSE
Lasso	-0.1322	0.1066	0.1696
Partial Least Squares	-1.2611	0.0953	0.2397
Bagged Splines	-0.1278	0.1027	0.1693

The results from this initial dataset were poor across the board. All three models produced negative out-of-sample (test) adjusted R^2 values, with Partial Least Squares performing worst at -1.2611, indicating the model's out-of-sample predictions were substantially worse than simply predicting the mean of the target variable.

More broadly, our models failed to capture the underlying variation in the data and the variance in the target variable. This was not a case of one model outperforming another. All three approaches, despite their different underlying mechanics, converged on a similarly poor result, which suggested the problem lay with the data itself rather than model choice.

Because the fundamentals-based predictors were already performing this poorly, we did not get to even testing the macro proxy variables (hyperscaler AI capital expenditure and regional drought monitors) at this stage.

In response, we expanded the stock universe to 28 tickers, adding more water-related and adjacent industrial stocks, to test whether a larger sample would improve model performance.

4. Significance of Predictors: 8-Stock Dataset

Predictor	Correlation	OLS Estimate	Std. Error	t-value	p-value	Significance ($p<0.05$)
vol_60d	-0.184	-0.2979	0.0598	-4.980	1.29e-6	Yes
div_yield	-0.044	-2.2869	0.6059	-3.774	2.07e-4	Yes
asset_growth	0.014	0.0537	0.0156	3.447	6.80e-4	Yes
ni_growth	0.159	0.0050	0.0019	2.675	0.0080	Yes
mom_12m	-0.128	-0.0858	0.0309	-2.779	0.0059	Yes
roe	-0.104	-0.8640	0.3600	-2.400	0.0172	Yes
debt_to_equity	-0.059	0.0252	0.0130	1.942	0.0534	No (Borderline)
current_ratio	0.025	0.0187	0.0104	1.797	0.0737	No
pb_ratio	-0.143	-0.0079	0.0052	-1.517	0.1306	No
mom_3m	-0.073	-0.0883	0.0573	-1.542	0.1246	No
capex_intensity	-0.066	-0.4857	0.3934	-1.235	0.2183	No
fcf_yield	0.033	0.3205	0.3237	0.990	0.3231	No
(Intercept)	-	0.1497	0.0510	2.932	0.0037	Yes

5. Results of Predictive Modelling on the 28-Ticker Dataset

Our expanded universe of 28 stocks included: AWK, WTRG, CWT, AWR, XYL, VRT, JCI, BMI, PNR, MWA, AOS, ITT, IEX, FELE, FLS, DOV, ECL, GRC, ITRI, MAS, ROP, EMR, PH, HON, WMS, GVA, PWR, and MTZ.

Model	Adjusted R^2	Train RMSE	Test RMSE
Lasso	-0.0928	0.1077	0.1364
Partial Least Squares	-0.1554	0.1064	0.1402
Bagged Splines	-0.0851	0.1061	0.1359

Expanding the universe to 28 tickers produced a clear improvement relative to the 8-stock results: test RMSE fell for every model, and the adjusted R^2 values, while still negative, moved substantially closer to zero. Partial Least Squares improved the most dramatically, from -1.2611 to -0.1554. Even so, none of the three models achieved a positive out-of-sample adjusted R^2, meaning the underlying issue had been reduced but not resolved.

6. The Pivot: Why We Moved Away from Our Initial Water-Stock Approach

Given that even the expanded 28-ticker water-sector dataset could not produce a positive out-of-sample R^2, we undertook a deeper diagnosis of what was actually limiting model performance, which led to a strategic pivot away from the water-stock-only approach.

Our initial water-stock universe was simply too small to give any model a workable number of observations relative to the number of predictors being used. With so few securities and time periods, the effective sample size was insufficient for the models to reliably learn stable relationships between predictors and returns.

Compounding the small-sample problem, the water stocks in our chosen universe tended to move together: they shared the same underlying drivers and the same macroeconomic exposures. This left very little independent variation across stocks for a model to actually learn from, since much of the cross-sectional information was redundant rather than incremental.

We concluded that more hyperparameter tuning or different feature engineering could not fix a dataset that fundamentally had too few observations and too little cross-sectional spread. The performance ceiling was set by the structure of the data itself, not by any deficiency in model specification.

This diagnosis was corroborated by existing academic research. The paper Empirical Asset Pricing via Machine Learning (2018, NBER) shows that even with a massive sample of nearly 30,000 stocks over 60 years, models still require enough usable data relative to the size of their predictor set in order to produce a stable, positive out-of-sample R^2. Our water-only universe fell far short of that bar, reinforcing that the path forward required a substantially larger and more diverse stock universe.

7. The Changes We Made

In response to the diagnosis above, we made three changes to our modelling approach.

Expanding the universe

We increased the stock universe to 222 from the initial 8. These stocks were all pulled from the S&P 500 with a full GICS sector spread, meaning stocks from varying industries not limited to a specific sector. This dramatically increased the variation present in the data and directly addressed the cross-sectional correlation problem that had constrained the water-only universe.

Shifting from fundamentals to technicals

After conducting significance testing, we also moved from mainly fundamental predictors to more technical predictors. Many fundamental predictors turned out to be statistically insignificant at the 95% confidence level while most technical predictors held up under testing. Our final predictor set remains a mix of both categories, but is now weighted much more heavily toward technical indicators than in the original model.

Updating the model shortlist

Through our research from the NEBR paper we also learned that our choice of models may also be posing an issue in terms of getting results. We moved from Lasso, PLS, and Bagged Splines to Gradient Boosted Regression Trees (GBRT), PLS, and Random Forest. We retained PLS since its earlier relative performance was decent, while Lasso and Bagged Splines were replaced with methods better suited to the revised data set, which was larger and more varied.

Updated predictor set

Category	Predictors

Technical / Risk	1-month momentum, 3-month momentum, 12-month momentum, 60-day volatility, dollar trading volume, 252-day beta, 60-day idiosyncratic volatility, % distance from 52-week high
Fundamentals (SEC)	Return on equity, debt-to-equity ratio, asset growth, net income growth, capex intensity
Price x Fundamentals	Price-to-book ratio, free cash flow yield

Model configurations

- GBRT: lambda = 0.01, Trees = 116
- Random Forest: Trees = 500
- PLS: Components = 1

8. Significance of Predictors: 222-Stock Dataset

Rank	Predictor	Relative Influence (%)
1	mom_12m	25.76
2	idio_vol_60d	16.64
3	pct_from_52w_high	16.62
4	beta_252d	10.91
5	mom_1m	7.35
6	vol_60d	4.19
7	pb_ratio	3.14
8	debt_to_equity	3.09
9	mom_3m	3.02
10	roe	2.62
11	fcf_yield	2.35
12	dollar_volume	1.82
13	asset_growth	1.56
14	capex_intensity	0.69
15	ni_growth	0.24

9. The Current Model

The current model framework reflects all tha changes discussed along with a more in-depth validation method.

We used walk-forward validation rather than standard k-fold cross-validation. This choice reflects the time-series nature of our data as standard k-fold cross-validation would cause future information to be part of our training dataset which would artificially improve performance and give inaccurate results. Walk-forward validation instead trains on past data and tests on subsequent dates in order to preserve chronology.

Our universe now spans 222 stocks, all sourced from the S&P 500, providing far greater cross-sectional breadth and sector diversity than the original water-only approach.

The complete-case rate was 84.1% (12,052 of 14,328 target-bearing rows) under the 15 predictors used consistently across all three models.

Our dependent variable is defined as excess return over Q1 and Q4 relative to the SPY benchmark. The underlying data consists of quarterly filing data spanning 2010 to 2026.

10. Results on Our Current Model

Model	Adjusted R^2	Test RMSE
GBRT	0.008395	0.1312
Partial Least Squares	0.01134	0.1310
Random Forests	-0.0175	0.1329

The table above represents the current model's performance. The expansion in the dataset, change in predictors and change in the models has produced positive walk-forward adjusted R^2 values for both GBRT and PLS compared to the 3 negative values from the earlier models. Random Forest remained slightly negative, suggesting it may be more prone to overfitting on this particular data structure relative to GBRT and PLS.

Test RMSE also improved substantially across the board. For example, PLS's test RMSE fell from 0.2397 on the original 8-stock dataset to 0.1310 on the current 222-stock dataset. This reduction of 45% further emphasizes how the changes we made improve predictive accuracy

11. Predictions for the Future

Using our current model, we generated forward-looking predictions by taking the top 30 best-predicted stocks and evaluating their projected performance relative to the S&P 500.

Projected performance

- Q3 2026: average 1.12% above the S&P 500, with average 4.49% growth
- Q2 2027: average 3.32% above the S&P 500, with average 18.26% growth

To illustrate the practical implication of these predictions, we calculated a simple hypothetical investment: if $10 were invested in each of the 30 stocks (a total of $300), the portfolio would be worth approximately $313.46 the following quarter, and approximately $354.79 the following year.

The 30 tickers underlying this prediction were: OKE, DVN, TRGP, BBY, HAL, OXY, CMG, DHI, PHM, NFLX, APTV, EOG, MCHP, MPC, FCX, NDAQ, COF, ADI, INTU, ADBE, NVR, EMR, SHW, TSCO, PNR, TMO, MS, MRSH, FITB, and VTR.

12. Top 5 Stocks Beating the S&P 500

Breaking down the top predicted performers by quarter provides further insight into which individual names are expected to drive the outperformance seen at the portfolio level.

Rank	Q3 2026	Q2 2027
1	MCHP - 3.67%	APTV - 6.03%
2	FCX - 3.24%	ADBE - 4.04%
3	APTV - 2.33%	COF - 3.83%
4	ADI - 2.13%	ADI - 3.57%
5	ADBE - 1.86%	TSCO - 2.22%

APTV and ADI appear in the top 5 for both quarters which suggests that the model consistently identifies these stocks as strong candidates for excess returns across different prediction horizons. ADBE also features prominently in both periods, moving from fifth place in Q3 2026 to second place in Q2 2027, indicating an improving predicted trajectory over the longer horizon. The magnitude of outperformance is also notably higher in the Q2 2027 projections (ranging from 2.22% to 6.03%) compared to Q3 2026 (1.86% to 3.67%). This is consistent with the higher average excess return reported at the portfolio level for the longer term prediction.